Pentagonal

ADVERSARIAL SMART CONTRACT REVIEW

Four.meme *template audit.*

Subject	Four.meme bonding-curve token template
Factory	0x5c95…0762b · TokenManager V2
Network	BNB Chain · chainId 56
Compiler	Solidity 0.8.x
Adversaries	8
Rules applied	2,184
Revision	Cached template · 2026-04-22

72
/ 100
MEDIUM RISK

0	2	3	4
CRITICAL	HIGH	MEDIUM	LOW / INFO

EXECUTIVE SUMMARY

What this template is, and what it isn't.

This is a **template audit**. Every token deployed by Four.meme's Token Manager shares an identical implementation. Pentagonal has already reviewed the template once; subsequent tokens inherit that review. We don't charge for re-running the red team against an artefact we've already characterised.

The template implements a standard bonding-curve memecoin with graduation to PancakeSwap once a market-cap threshold is crossed. It is not malicious by design, but it ships with the same operational risks every memecoin launchpad template carries: the deployer's wallet can disproportionately influence early price discovery, taxes are mutable until graduation, and the token's economic identity is entirely cosmetic — name, symbol, and image only.

The template **passes** four of the eight adversaries (no reentrancy, no oracle dependence, no overflow risk, no MEV-trivial swap path on the bonding-curve side). It **flags** for the remaining four — detailed below.

Findings overview

SEVERITY	COUNT	NOTES
CRITICAL	0	No drain vector exists in the template itself.
HIGH	2	Deployer-wallet concentration, mutable trading parameters.
MEDIUM	3	Graduation race, sandwich exposure on graduated pair, tax window.
LOW / INFO	4	Standard ERC-20 with documented quality-of-life additions.
Total	**9**	—

Adversary scorecard

ADVERSARY	STATUS	FINDINGS
Reentrancy Hunter	✓ Clear	0
Flash Loan Attacker	✓ Clear	0
Access Control Prober	⚠ Findings	1 (high)
Overflow Saboteur	✓ Clear	0
Oracle Manipulator	✓ Clear	0
MEV Predator	⚠ Findings	1 (medium)
Economic Exploit	⚠ Findings	4
Gas Griefer	⚠ Findings	1 (info)

HIGH FINDINGS

What's worth knowing before you buy.

F-01 · Deployer wallet receives concentrated allocation pre-graduation

HIGH

Found by `Economic Exploit`

The Four.meme template allocates the bulk of pre-graduation supply to the bonding curve, but the deployer is permitted to buy in early at the lowest curve price. In practice this means the deployer typically holds 10–30% of post-graduation supply, frequently more when they execute a coordinated initial buy.

> **What it means for you.** Treat any chart on a Four.meme token as trading against a single concentrated holder until you can verify their wallet has been distributed or burned post-graduation. Liquidity depth at graduation is a *snapshot*, not a guarantee.

F-02 · Trading parameters mutable until graduation

HIGH

Found by `Access Control Prober`

Buy and sell tax, max-transaction limits, and max-wallet limits can be modified by the deployer while the token sits on the bonding curve. Common patterns observed in the wild:

- Tax bumped to 99% on sell to lock the chart artificially
- Max-wallet reduced to disable buying past a target market cap
- Max-tx tightened to grief sniping bots

These parameters typically renounce on graduation, but **not always** — some templates retain admin controls on the post-graduation pair through an external taxer contract.

> **Remediation guidance.** The template is what it is. Any user-facing copy on a Four.meme token's resale risk should explicitly call out that admin controls remain live during the bonding-curve phase.

MEDIUM FINDINGS

Operational and economic edges.

F-03 · Graduation produces sandwich exposure on first PancakeSwap block

MEDIUM

Found by MEV Predator

The instant a Four.meme token graduates to PancakeSwap, the first block of trades on the new pair is a known MEV target. Bots monitor the Token Manager's graduation event and sandwich the liquidity addition. Retail buyers in the first block typically eat 1–5% slippage to MEV alone.

F-04 · Tax-modification window during the bonding-curve phase

MEDIUM

Found by Economic Exploit

Same root cause as F-02; classified separately because the *economic impact* is distinct from the *access surface*. A tax change between your buy and your sell can change your exit price by an order of magnitude. This risk is binary: either it happens to you or it doesn't, and you have no on-chain warning.

F-05 · Bonding-curve liquidity is the manager contract's, not yours

MEDIUM

Found by Economic Exploit

While a token is on the bonding curve, the liquidity backing your position lives inside the Four.meme Token Manager contract, not in a pair you control. If the manager is paused, upgraded, or deprecated, your ability to exit is gated on the platform. Graduation transfers liquidity to PancakeSwap and removes this dependency.

LOW / INFORMATIONAL

Background notes.

- **F-06** · Standard ERC-20 with anti-bot toggles. No surprises.
- **F-07** · No reentrancy vectors in the template.
- **F-08** · Arithmetic is checked throughout; no unchecked blocks of concern.
- **F-09** · Manager contract holds substantial BNB balance — not a concentration risk for individual tokens, but a platform-level concern for users tracking systemic exposure.

RECOMMENDATION

How to size and where to use this template.

The Four.meme template is **technically sound** at the contract level. It does not contain a critical drain vector, a reentrancy exposure, or an oracle-manipulation surface. The risks it carries are **economic and operational**, not exploit-class:

1. The deployer holds disproportionate pre-graduation supply.
2. Trading parameters are mutable until graduation.
3. Graduation is a known MEV target.
4. While on the curve, your exit depends on the platform.

For **short-duration speculative exposure**, the template is acceptable provided the user understands these constraints. For **long-term holding** — treasury allocation, savings, payment rails — Four.meme tokens are inappropriate. Recommend users size positions accordingly and prefer post-graduation entries with verified deployer-wallet behaviour.

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